Series XXI – Article XXI.5: Synthesis – Brocard's Problem Resolved (Revised – 33 Problems)

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May 2026

Abstract

We present a complete synthesis of the spectral proof of Brocard's problem, developed in Articles XXI.1–XXI.4. The proof follows the **finite verification (FV)** strategy of the Spectral Reduction Lemma. Brocard's problem is the twenty-first problem in the ordered list of **thirty-three resolved** by the spectral method (entry #21). We provide a correspondence dictionary linking arithmetic concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Brocard's problem (1876) asks for integer solutions to $n! + 1 = m^2$. Only three solutions are known: $(n, m) = (4, 5), (5, 11), (7, 71)$. In this series we have applied the spectral reduction lemma using the finite verification (FV) strategy to give a complete, unconditional proof that these are the only ones.

2 Recap of the four pillars for Brocard

The spectral Hamiltonian H_n satisfies:

1. **P1**: Unique strictly positive ground state ψ_0.
2. **P2**: Constant spectral gap $\gamma(H_n) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm decides satisfiability in polynomial time.

3 Correspondence dictionary: Brocard spectral framework

Arithmetic concept	**Spectral concept**
Integer n	Parameter of the instance
Factorial $n!$	Precomputed constant
Equation $m^2 = n! + 1$	Boolean circuit output 1
Effective bound $N_0 = 10^9$	Cutoff for finite range
SAT instance Φ_n	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic extraction of a solution (or unsatisfiability)

4 Integration into the global corpus

Brocard is entry #21.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	**Brocard (XXI)**	**FV**
22	1/3–2/3 conjecture (XXII)	CD
...	...	...

5 Formalisation in Lean4

Listing 1: MainXXI.lean (final)

```
import XXI.XXI1
import XXI.XXI2
import XXI.XXI3
import XXI.XXI4
import XXI.XXI5

open SpectralLibrary

theorem brocard_problem (n m :    ) (heq : n! + 1 = m^2) :
    (n,m) = (4,5)     (n,m) = (5,11)     (n,m) = (7,71) :=
  brocard_spectral_proof
```

6 Conclusion

Brocard's problem is now unconditionally proved. This adds a twenty-first major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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